Jonathan Poteet

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Professional Summary

Software Engineer with 5+ years building and integrating production-grade enterprise applications. Contributed to GM's platform modernization by connecting services, APIs, and data across applications while improving system reliability and secure delivery.

Technical Skills

Frontend and Web: Angular, Solid.js, React, Next.js, TypeScript, JavaScript, HTML, CSS/SCSS
Backend and APIs: Node.js, Spring Boot, Kafka, Java, Python, REST APIs, SQL
Databases: MongoDB, MySQL, PostgreSQL
Cloud, DevOps, and Tools: Kubernetes, Docker, Ansible, Azure DevOps, CI/CD
Testing and Security: Vitest, GitHub Advanced Security (GHAS)
Systems and Networking: Linux, WSL, TCP/IP

Professional Experience

Pod Juggler LLC September 2026 — Present

Owner and Software Developer — Remote

- Founded Pod Juggler LLC to pursue software consulting and independent development; built small-scale software projects while preparing for potential client work.

General Motors March 2021 — December 2025

Software Engineer — Roswell, GA

- Owned the My Orders frontend experience from checkout through delivery, onboarding the website in Sweden, France, Germany, and Switzerland. Built with Next.js, React, and TypeScript while integrating Strapi CMS and enterprise APIs for localized vehicle and financial data.
- Supported production rotations and resolved application issues across My Orders, using monitoring and analytics to investigate incidents and help maintain service reliability.
- Onboarded data services and APIs, including UVeye and the SAVI dongle, into the Service Workbench platform; debugged integrations and clarified requirements with cross-functional teams.
- Contributed to modernization of the enterprise technician platform by re-engineering legacy architecture with Angular, TypeScript, and Java to improve scalability, maintainability, and responsiveness.
- Built Docker and Podman development environments with WSL for multi-repository Linux microservices, reducing setup friction and deployment errors.
- Developed REST APIs and backend services with Java and Spring Boot, and managed Azure DevOps CI/CD pipelines to automate software delivery.
- Led migration of the My Orders platform to Solid.js in the DASH monorepo and introduced Vitest testing to strengthen frontend quality; remediated GitHub Advanced Security findings and monitored Kubernetes pods supporting microfrontend services.
- Mentored two to four developers in frontend development, sharing implementation guidance and supporting their growth on the team.

Education

Vanderbilt University December 2025
Master of Science in Computer Science — GPA: 3.97

Florida State University December 2020
Bachelor of Science in Information Technology — Minor in Business — GPA: 3.80

Certifications

CompTIA Security+ (SY0-701) 2026
Global Citizenship Certificate — Florida State University 2020

Selected Projects

Sinkhole — Procedural Dungeon Crawler (Godot/C#/.NET) Jan 2026 – Present

- **Infinite World Architecture:** Engineered a chunk-based streaming system in Godot 4 (C#/.NET) that generates world data at runtime using seeded random algorithms.
- **Procedural Generation:** Developed custom layout logic to construct non-linear, endlessly descending floors, balancing randomness with navigable structural constraints.
- **Performance Optimization:** Decoupled generation logic from the main render thread to prevent frame-rate stutters during high-intensity procedural generation cycles.

World State AI — Resource Optimization Simulation (Python/LISP) Aug 2025 – Oct 2025

- **Expected Utility Algorithm:** Engineered an incremental algorithm to evaluate and rank action schedules based on calculated expected utility and reward functions.
- **Decision Logic:** Implemented a parser to translate complex state transitions into executable schedules, identifying optimal resource trades to maximize global state stability.
- **Scalable Modeling:** Developed weights for diverse resource types, enabling the simulation of complex international trade and transformation resource loops.

Probabilistic Sensor Verification (Python/PRISM) Oct 2025 – Nov 2025

- **Formal Verification & Modeling:** Developed a framework to evaluate sensor reliability using PRISM model checker, modeling gas/thermal sensors as DTMC to simulate failure states.
- **Safety & Risk Assessment:** Authored PCTL safety properties to analyze system reachability and hazard thresholds, identifying theoretical failure points and safety boundaries.
- **Statistical Analysis:** Conducted cross-validation of formal results against Monte Carlo simulations using StormPy, evaluating variance between mathematical methods and empirical simulation.

Kubernetes Cluster for Distributed Systems (K8s/Python) Oct 2024 – Nov 2024

- **Cloud Orchestration:** Architected a K8s cluster across four VMs on Chameleon Cloud. Resolved complex node-to-node networking and resource contention challenges.
- **Distributed Systems:** Integrated Apache Kafka for high-throughput messaging, optimizing resource provisioning to scale throughput by 10x (up to 5,000 messages per node).
- **Automation:** Authored custom Ansible playbooks and designed Dockerfiles to ensure environment parity and enforced deployment immutability via CI/CD.

Professional Portfolio — jonathanpoteet.com

- **React + Next.js:** Built with React and Next.js and deployed via GitHub Pages to showcase modern frontend architecture, component-driven development, and scalable web experiences.
- **Creative Engineering:** Combines formal software engineering with procedural generation and 3D design (Three.js, Blender) to build interactive, logic-driven environments.
- **Design-to-Code Execution:** Translates visual design concepts into polished frontend experiences with careful attention to structure, interaction, and implementation quality.

Campus and Community Involvement

Toastmasters	2026 — Present
Study Abroad: FSU International Programs — London, England	Summer 2019
Air Force ROTC (AFROTC) — Detachment 145	2017 — 2018
Florida American Legion Boys State — Delegate	2016
Eagle Scout, Boy Scouts of America	2016